CS 5600
Fall 2026
Assignment 0 – Python Fundamentals
Due: 08/28/2026 11:59 pm

The goal of this assignment is to refresh the Python programming and scientific computing skills that we will use throughout this course. In particular, you will practice:

- Basic Python programming
- NumPy arrays and vectorized operations
- Data visualization with Matplotlib
- Writing and formatting text using Markdown cells in a Jupyter notebook
- Interpreting numerical and graphical results

This assignment contains **four problems, each worth 5 points**, for a total of **20 points**. Each problem will be graded as follows: code segments are worth 4 points and the accompanying text description will be worth 1 point.

Please submit your work as a single *Jupyter/IPython notebook (.ipynb)*. Use *Google Colab* to create and test your notebook, as submissions will be graded in Colab to ensure a common execution environment. You can export it as "File → Download → Download .ipynb" for submission.

Your notebook should:

- Run from beginning to end without errors using *Runtime → Run all* in Google Colab.
- Include all required import statements.
- Use clear variable names and brief comments where appropriate.
- Use *code cells* for programming tasks.
- Use *text/Markdown cells* for all written-response questions.
- Print or display all requested results.
- Generate the requested figures and save them using the exact filenames specified below.

Do not submit generated PNG or PDF files separately. They will be generated when your notebook is executed.

Problem 1: Python Programming Basics — 5 points
In this problem, you will review basic Python data structures, functions, and control flow.

Part A: Lists and Dictionaries

Create a Python list containing the following five hypothetical model accuracies:

`0.72, 0.81, 0.79, 0.88, 0.84`

Using Python, print:

1. The first accuracy
2. The last accuracy
3. The number of values in the list
4. The largest accuracy
5. The average accuracy

Next, create a dictionary with the following information:

```
model_name: "Baseline Classifier"
accuracy: 0.84
num_features: 12
```

Print each value from the dictionary.

Part B: Functions

Write a Python function named `classify_score(score)` that:

- Returns "`high`" if the score is greater than or equal to `0.85`
- Returns "`medium`" if the score is greater than or equal to `0.75` but less than `0.85`
- Returns "`low`" otherwise

Call the function on every value in the accuracy list and print the results.

Part C: Written Response

Create a *Markdown/text* cell and answer the following in 2–4 sentences:

What is the difference between a Python list and a dictionary? Give one example of when each might be useful when working with data in an AI application.

Problem 2: NumPy Arrays and Operations — 5 points

In this problem, you will practice creating, manipulating, and performing computations with NumPy arrays.

1. Import NumPy.
2. Set the NumPy random seed to 42 so that your results are reproducible.
3. Create a *1D NumPy array containing 5 random values* drawn from a uniform distribution between 0 and 1.
4. Print:
    - The array
    - The shape of the array
5. Compute and print:
    - The mean
    - The standard deviation
    - The minimum value
    - The maximum value
6. Reshape the array into a *2D array with 5 rows and 1 column*.
7. Print:
    - The reshaped array
    - Its new shape
8. Add 5 to every element of the reshaped array using a NumPy operation. Print the result.
9. Compute and print the *dot product of the original 1D array with itself.*
10. Use Boolean indexing to print only the values in the original array that are greater than 0.5.

Do not use Python loops for the NumPy computations above.

Written Response

Create a *Markdown/text cell* and answer the following:

The original array has shape (5,), while the reshaped array has shape (5, 1). In 1–3 sentences, explain what is different about these two shapes.

You do not need to provide a formal mathematical definition. Explain the distinction in your own words.

Problem 3: Plotting with Matplotlib — 5 points

In this problem, you will practice generating numerical data and visualizing it with NumPy and Matplotlib.

1. Import Matplotlib's `pyplot` module.
2. Create an array of x values ranging from 0 to 10 with an increment of 0.1.
3. Using NumPy, compute:
$$y_1 = \sin(x)$$
$$y_2 = \cos(x)$$
4. Plot both functions on the *same figure.*
5. Your figure must include:
    - A label for the x-axis
    - A label for the y-axis
    - A descriptive title
    - A legend identifying the sine and cosine curves
    - Gridlines
6. Use different line styles for the two curves.
7. Save the figure as: `trig_functions.png`
8. Display the figure in the notebook.

Use NumPy operations to calculate the function values rather than Python loops.

Written Response

Below your figure, create a *Markdown/text cell* and answer the following in 2–4 sentences:

Describe one similarity and one difference between the sine and cosine curves shown in your plot.

Your response should refer to something that is visible in the graph.

Problem 4: NumPy and Matplotlib for Data Analysis — 5 points

In this problem, you will combine NumPy array operations, indexing, summary statistics, and Matplotlib visualization.

1. Create a NumPy array containing x values from 0 through 100, inclusive, with an increment of 1.
2. Compute the corresponding values of the quadratic function

$$y = 0.5x^2 + 2x + 1$$

using NumPy array operations.

3. Print:
    - The first 5 values of x
    - The first 5 values of y
    - The shape of x
    - The shape of y
4. Find and print the value of y corresponding to x = 50.
5. Compute and print the mean of all values in y.
6. Create a plot of the quadratic function.
7. Your plot must include:
    - Appropriate x- and y-axis labels
    - A descriptive title
    - A legend
    - Gridlines
    - A line style of your choice
8. Mark the point corresponding to x = 50 using a scatter point.
9. Save the figure as: `quadratic_function.pdf`
10. Display the figure in the notebook.

Use NumPy operations whenever possible rather than explicitly looping through the arrays.

Written Response

Create a *Markdown/text cell* below the plot and answer the following in 2–4 sentences:

How does the value of (y) change as (x) becomes larger? Based on the equation and your graph, why does the graph have this shape?

Before submitting your notebook, verify that:

1. The notebook runs from beginning to end without errors in Google Colab.
2. You used Markdown/text cells for all written responses.
3. Your notebook includes clear section headings.
4. All requested numerical values are printed.
5. Your plots contain the requested titles, labels, legends, and gridlines.
6. Your notebook generates `trig_functions.png`.
7. Your notebook generates `quadratic_function.pdf`.
8. You did not submit the generated PNG or PDF files separately.
9. You submitted a single .ipynb file.